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HORIZON

**AI-Based Virtual Camera Tracking System for
Coarse Alignment of Mobile Free Space Optical
Communication (FSOC) Terminals**

Technical Report

Problem Statement ID:	SIH26169
System Designation:	HORIZON Coarse Alignment & PAT Suite
Report Scope:	Problem Understanding, Architecture, Tracking, AI, Verification and Evaluation
Report Version:	1.0 — Expanded Technical Edition
Document Type:	Technical Report

Technical report distilled from the full 12-section dossier, retaining the highest-value engineering content, equations, architectures, verification evidence, benchmark metrics, failure analysis and roadmap.

Document Control

Field	Value
Document	HORIZON Technical Report
Problem Statement	SIH26169
System	HORIZON Coarse Alignment & PAT Suite
Version	1.0 — Curated 20-Page Edition
Status	Research-Grade Technical Report
Primary Scope	Problem, architecture, modelling, AI, tracking and evaluation
Platform	HORIZON AI-Based Virtual Camera Tracking System

Technical Note

This edition intentionally preserves the original 11pt typography and report visual language. Content has been shortened by selecting the most decision-relevant technical material rather than reducing font size.

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HORIZON TECHNICAL REPORT

1 Problem Understanding & Operational Context

1.1 The FSOC Spatial Acquisition Dilemma

FSOC uses extremely narrow optical beams. The diffraction-limited divergence is governed by aperture and wavelength:

$$\theta_{\text{div}} \approx 1.22 \frac{\lambda}{D}. \quad (1)$$

The source specifies typical aerospace apertures $D \in [10, 30]$ cm and a practical divergence envelope of $5\text{--}50\ \mu\text{rad}$. Initial pointing uncertainty can span $1^\circ\text{--}5^\circ$, while the fine QPD/FSM capture region is only $500\text{--}2000\ \mu\text{rad}$. For the supplied reference case,

$$P_{\text{acquisition}} \approx \left(\frac{10\ \mu\text{rad}}{17,500\ \mu\text{rad}} \right)^2 = 3.26 \times 10^{-7}. \quad (2)$$

The virtual camera therefore bridges the wide uncertainty cone and the narrow fine-pointing basin by estimating (u, v) and steering until $e \leq 3\ \text{px}$.

1.1.1 Capture-Basin Scale Separation

The supplied operating picture contains a deliberate scale mismatch: the initial uncertainty cone is measured in degrees, the coarse camera works in pixels over a degree-scale field, and the final QPD/FSM handoff is measured in hundreds of microradians. HORIZON therefore evaluates coarse alignment as a problem of keeping the optical beacon inside a shrinking capture basin rather than as a one-shot pointing command.

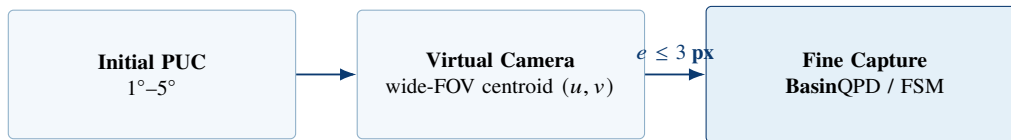


Figure 1: Geometric scale separation between initial pointing uncertainty, virtual-camera acquisition, and fine optical capture.

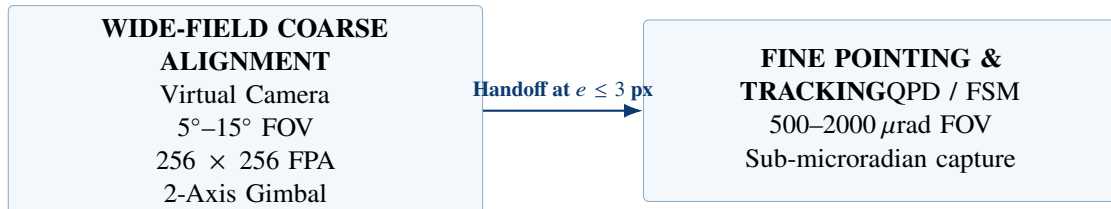


Figure 2: Coarse-to-fine PAT hierarchy.

1.2 Physical Degradation Taxonomy

The full source identifies six concurrent stress vectors: platform jitter, atmospheric turbulence/scintillation, FPA sensor noise, impulse noise, dynamic target maneuvers, and optical distractors. Their critical

envelopes are summarized below.

Table 1: Core disturbance envelope.

Stress	Supplied envelope	Primary effect
Platform jitter	1–50 Hz; resonances 5.0, 12.5, 27.8 Hz; up to ± 20 px/frame	Centroid motion and phase lag
Atmosphere	C_n^2 from 10^{-17} to 10^{-12}	Beam wander and fading
FPA noise	Poisson + Gaussian readout	Low-SNR centroid degradation
Impulse noise	0 / 255 DN events	Beacon-like false points
Target maneuvers	Up to 15g	Fixed-CV model mismatch
Optical clutter	Sun, albedo, clouds, debris	False candidates / false lock

1.3 SIH26169 Engineering Targets

The principal release gates are 100 Hz execution, $e_{95} \leq 3$ px, nominal $e_{95} \leq 1$ px, complete PAT recovery, operation to the supplied 0 dB gate, physical gimbal limits and a 1,016-test verification suite.

1.4 Architectural Ground-Truth Firewall

Perception receives only noisy image arrays; estimation receives noisy 2D centroids and innovation data; PAT uses statistical evidence; the true state is reserved for the external recorder. This separation is the central validity invariant of the simulator.

2 End-to-End System Architecture & Data Flow

2.1 Two Structurally Isolated Domains

The physical domain contains the deterministic 100 Hz clock, target dynamics, optical projection, disturbance stages and FPA synthesis. The autonomous domain contains perception, estimation, PAT supervision and control. The firewall exposes only the corrupted sensor frame and gimbal encoder telemetry.

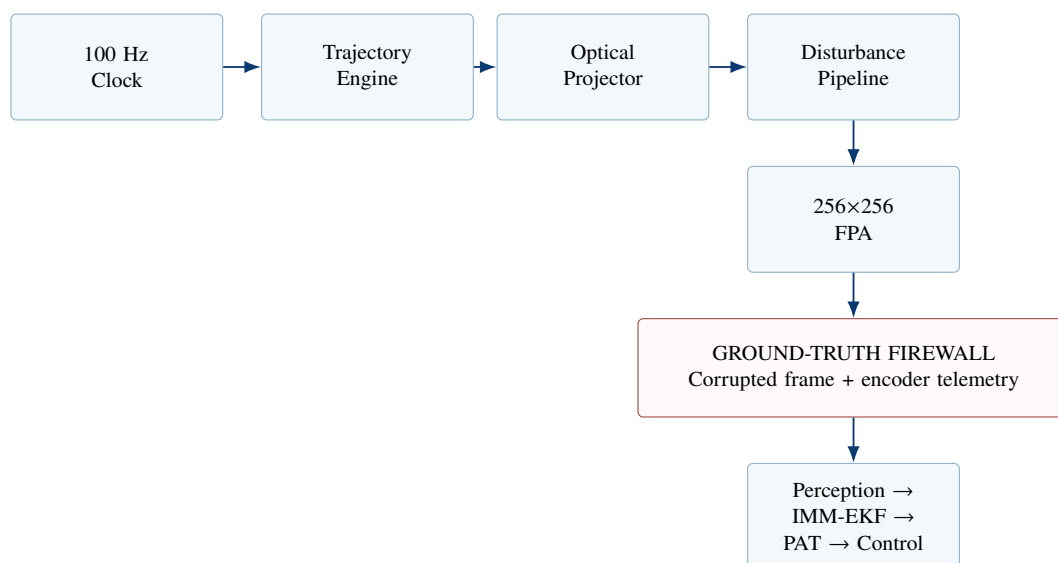


Figure 3: End-to-end HORIZON architecture.

2.2 Zero-Leakage Information Constraint

With true state $\mathbf{X}_{\text{GT}}$, observed stream $\mathbf{Z}_{\text{obs}}$ and encoder history θ_{enc} , the source formalizes the information constraint as

$$I(\mathbf{u}_{\text{cmd}}(t); \mathbf{X}_{\text{GT}}(t) \mid \mathbf{Z}_{\text{obs}}^{0:t}, \theta_{\text{enc}}^{0:t}) \equiv 0. \quad (3)$$

The implementation enforces this through module-import isolation, explicit interface typing and the leakage test suite.

2.3 Deterministic 100 Hz Execution Cycle

The complete tick executes ten stages: kinematic propagation; camera-frame transformation; disturbance injection; firewall crossing; candidate perception; IMM-EKF estimation; PAT state execution; controller computation; actuator rate/acceleration limiting; and telemetry ingestion.

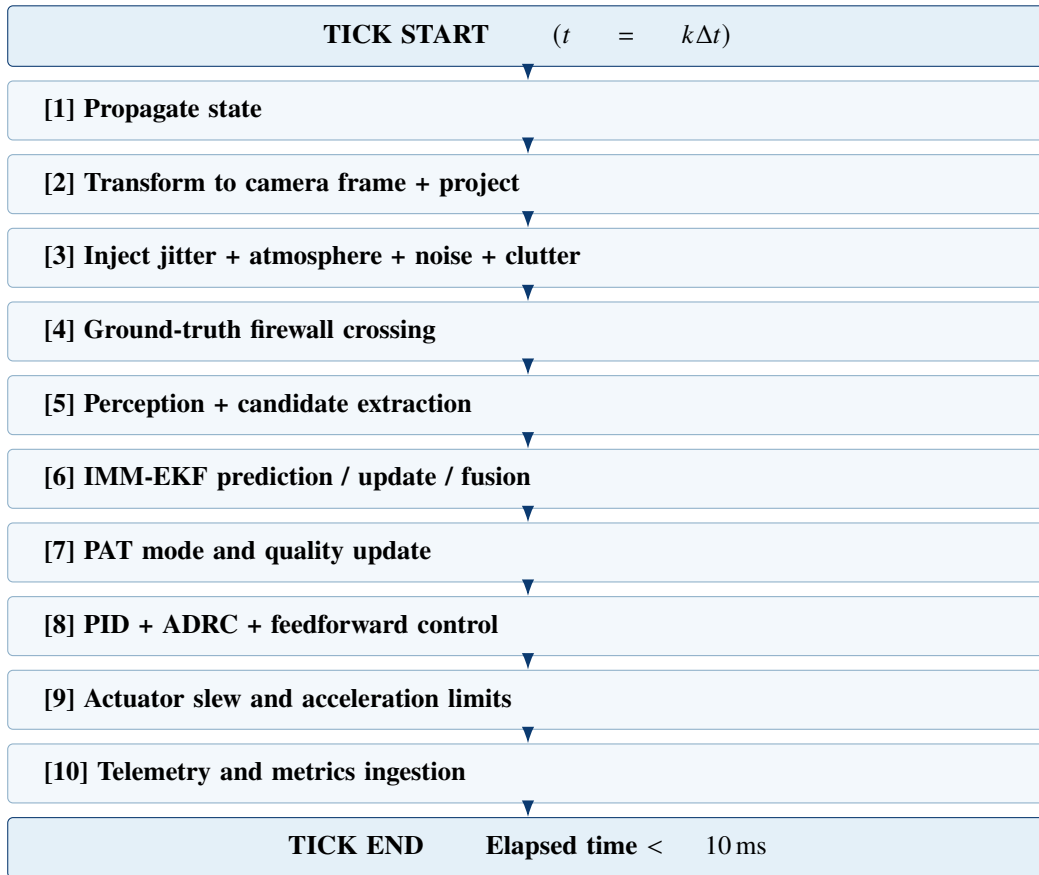


Figure 4: Deterministic execution sequence within the 100 Hz master cycle.

2.4 Hardware-in-the-Loop Video Injection

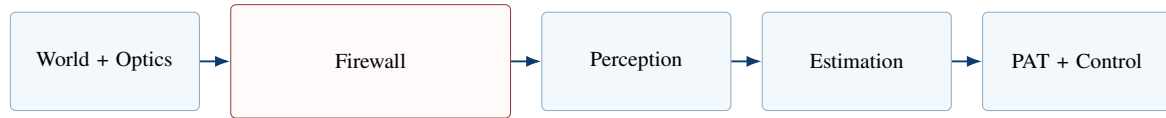
The HIL interface supports MP4/AVI or CSV inputs, aligns timestamps with a monotonic timebase and normalizes 8/10/12-bit sources to the simulator's 8-bit DN range without altering the internal tracking logic.

2.5 Interface Contracts and Observability Boundary

The architecture maintains a narrow observable interface between the physical simulation and the autonomous payload. The contract design makes the payload auditable because every downstream decision can be traced to the image, encoder history and previously estimated state available at the firewall.

Table 2: Primary data contracts across the HORIZON processing chain.

Boundary	Exposed information	Receiving stage
Sensor boundary	Corrupted 256×256 frame + gimbal encoder telemetry	Perception
Detection contract	Centroid, confidence, validity, SNR, source	IMM-EKF
State contract	Six-state estimate, covariance, model probabilities	PAT / control
Command contract	Bounded pan/tilt rate command	Gimbal model
Evaluation channel	Ground truth, error and mode history	External recorder only

**Figure 5:** Observable data path across the HORIZON architecture.

3 Software Modular Decomposition & Subsystems

3.1 Architectural Modular Decomposition

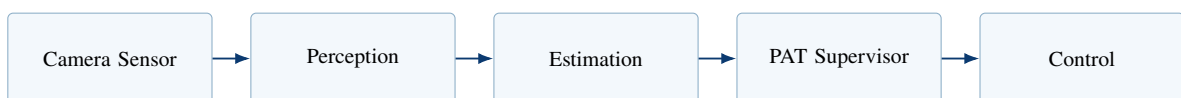
HORIZON is implemented in Python 3.11+ under `coarse-align-x/`. Major packages include `simulator/`, `estimation/advanced/`, `control/`, `pat/`, `benchmarks/` and `tests/`; core files include `main.py` and `configs/default.yaml`.

Table 3: Subsystem responsibilities.

Subsystem	Core responsibility
Core	100 Hz clock, configuration, seed management, ground-truth recorder
World / Trajectory	Six analytical trajectories including RandomMotion
Camera / Optics	Pinhole model, Brown–Conrady distortion, PSF, gimbal limits
Disturbances	Jitter, turbulence, scintillation, noise, impulse and distractors
Perception	Classical morphology, CoG, parabolic fit, YOLOv8n, hybrid fusion
Estimation	6-DOF IMM-EKF with CV, CA and MAN models
Control	PID, ADRC, Smith prediction, feedforward and camera controller
PAT	Five-state supervisory lifecycle

3.2 Inter-Subsystem Data Contracts

The principal contracts are `DetectionResult`, `StateEstimate`, PAT state and bounded command-rate objects. These interfaces contain only the information needed by the receiving stage, making the module boundaries testable and auditable.

**Figure 6:** Explicit data-contract chain across the autonomous payload.

3.3 Software-Level Isolation

The source explicitly separates the physical world engine from perception, estimation, control and PAT modules. This isolation supports reproducibility, prevents truth leakage and permits focused component-level verification.

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The principal repository structure is `coarse-align-x/`, with deterministic simulation under `simulator/`, advanced estimation under `estimation/advanced/`, control under `control/`, supervisory logic under `pat/`, and dedicated `benchmarks/` and `tests/` packages. Core files include `main.py` and `configs/default.yaml`.

3.4 Module Dependency Map

The repository is decomposed so that high-coupling physical components remain upstream of the firewall, while autonomous modules form a downstream chain of typed interfaces. This structure is consistent with the requirement that perception, estimation and control operate on sensor observables rather than simulator truth.

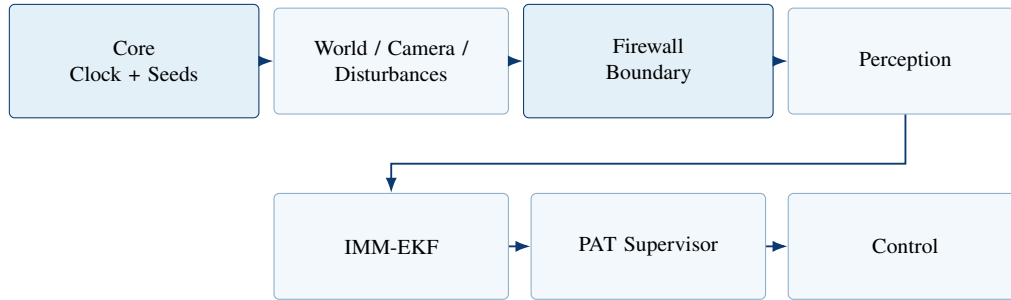


Figure 7: Module dependency map showing the physical-to-autonomous execution boundary.

4 Physical Environment & Sensor Degradation Modeling

4.1 Deterministic Simulation Engine & Seed Architecture

The simulation advances at 100 Hz with $\Delta t = 0.01$ s. A root seed is domain-separated into subsystem-specific random streams using SHA-256 so that trajectory, jitter, atmosphere, noise and distractors remain reproducible and independently testable.

$$S_{\text{sub}} = \text{uint32}(\text{SHA256}(S_0 \parallel \text{subsystem_identifier})[0:4]) \bmod 2^{32}. \quad (4)$$

4.2 Multi-Tone Structural Platform Jitter Model

The jitter model combines structural resonances at 5.0, 12.5 and 27.8 Hz with colored Gauss–Markov noise. The supplied envelope reaches ± 20 px/frame at 100 Hz.

$$J(t) = \sum_i A_i \sin(2\pi f_i t + \phi_i) + n_{\text{GM}}(t). \quad (5)$$

4.3 Atmospheric Turbulence and Scintillation Physics

The refractive-index field is written as $n = n_0 + \delta n$ and characterized by C_n^2 . The supplied Rytov approximation is [2, 4]

$$\sigma_\chi^2 = 0.563 C_n^2 k^{7/6} L^{11/6}, \quad I = I_0 \exp(2\chi), \quad \chi \sim \mathcal{N}(-\sigma_\chi^2, \sigma_\chi^2). \quad (6)$$

Beam wander is represented by the supplied variance form $0.97 C_n^2 L^3 W_0^{-1/3}$, while the scintillation index is $\exp(4\sigma_\chi^2) - 1$.

4.4 Composite FPA Sensor Degradation

The sensor model combines Gaussian PSF, Poisson shot noise, Gaussian readout/thermal noise and impulse corruption. The source specifies an approximately 1–2.5 px PSF envelope, 2–15 DN readout noise and impulse probability 0.001–0.05.

$$I_{\text{FPA}} = Q[I_{\text{PSF}} + I_{\text{clutter}} + n_{\text{shot}} + n_{\text{read}} + n_{\text{impulse}}]. \quad (7)$$

Table 4: Physical link scenarios used in the full evaluation envelope.

Link	Range / altitude	C_n^2 regime
LEO–Ground	500–1500 km; 0–500 km atmosphere	10^{-15} – 10^{-13} severe envelope
UAV–Ground	10–50 km; 0–10 km	5×10^{-14} – 10^{-12} moderate/severe
HAPS–HAPS	50–200 km; 18–25 km	10^{-17} – 10^{-16} weak
GEO OISL	36,000 km; vacuum	0

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4.5 Disturbance Composition and Sensor Synthesis

The disturbance stack is applied sequentially before the firewall. Structural jitter perturbs apparent centroid location; atmospheric turbulence contributes scintillation and beam wander; optical clutter introduces competing sources; and sensor statistics convert the disturbed optical field into the 8-bit measurement presented to perception.

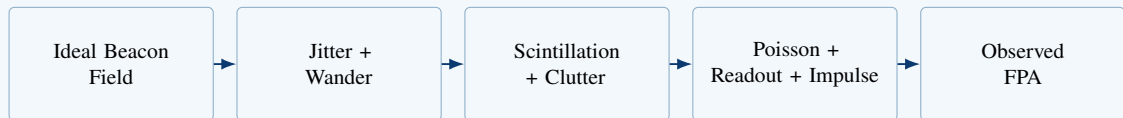


Figure 8: Sequential optical-to-sensor disturbance synthesis.

For the PSF representation, the source uses a compact Gaussian envelope with approximately 1–2.5 px width. The FPA synthesis then combines the optical signal, clutter and noise before quantization to the documented 8-bit dynamic range. This separation lets the same corrupted frame interface feed multiple perception implementations without changing the physical disturbance generator.

Table 5: Key physical degradation parameters retained from the full model.

Parameter	Supplied range	Role in simulation
Structural jitter frequency	1–50 Hz; peaks at 5.0, 12.5, 27.8 Hz	High-rate centroid motion
Jitter displacement	Up to ± 20 px/frame	100 Hz tracking challenge
Beam wander	$0.97C_n^2 L^3 W_0^{-1/3}$	Random spot displacement
PSF width	$\sigma \approx 1\text{--}2.5$ px	Optical blur representation
Readout noise	2–15 DN	Sensor/electronics noise
Impulse probability	0.001–0.05	Salt-and-pepper corruption
Distractors	1–5 moving sources	False-candidate generation

5 Classical Perception & Sub-Pixel Optics

5.1 Optical Preprocessing and Background Suppression

The classical chain is Gaussian 3×3 smoothing, white Top-Hat filtering with a 5 px disk, dynamic Otsu thresholding, 8-connected components, geometric filtering and sub-pixel localization.

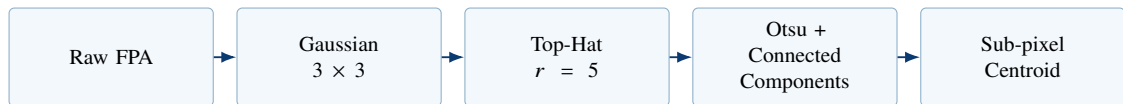
The Gaussian stage suppresses high-frequency sensor noise while preserving compact optical spots:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left[-\frac{x^2 + y^2}{2\sigma^2}\right]. \quad (8)$$

The white Top-Hat transform is $T = f - (f \circ B)$, where B is the disk-shaped structuring element. Otsu thresholding then selects the intensity threshold maximizing inter-class variance,

$$\sigma_B^2(t) = \omega_0(t)\omega_1(t) [\mu_0(t) - \mu_1(t)]^2. \quad (9)$$

The classical chain is Gaussian 3×3 smoothing, white Top-Hat filtering with a 5 px disk, dynamic Otsu thresholding, 8-connected components, geometric filtering and sub-pixel localization.

**Figure 9:** Classical deterministic beacon-perception pipeline.

5.2 Candidate Filtering and Sub-Pixel Centroiding

Candidate regions are retained within area $A \in [3, 150]$ px, circularity $4\pi A/P^2 \geq 0.35$ and an SNR threshold computed from a local annular neighbourhood. Two source localizers complement each other: squared CoG is the fast path, while the three-point Gaussian/parabolic peak fit is used when the local spot shape supports it.

$$u_{\text{CoG}} = \frac{\sum_i w_i^2 u_i}{\sum_i w_i^2}, \quad v_{\text{CoG}} = \frac{\sum_i w_i^2 v_i}{\sum_i w_i^2}. \quad (10)$$

The source's precision envelope spans approximately 0.015, 0.052 and 0.118 px for 10,000, 1,000 and 250 electrons under progressively weaker observation conditions.

5.2.1 Centroiding Decision Envelope

The fast squared-CoG estimator and the parabolic peak fit are complementary rather than mutually exclusive. The supplied implementation uses local image evidence and source conditions to select the refinement path while retaining the same centroid interface for downstream estimation.

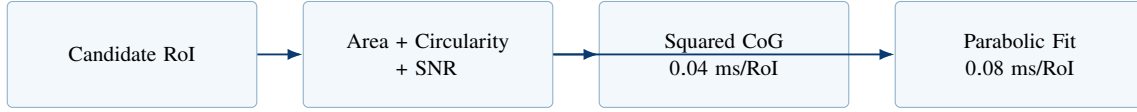


Figure 10: Classical candidate validation and sub-pixel refinement options.

5.3 Cramér–Rao Lower Bound

The localization bound is expressed through Fisher information:

$$\text{Var}(\hat{u}) \geq \mathcal{I}(u)^{-1}, \quad \mathcal{I}(u) = \sum_i \frac{1}{\lambda_i} \left(\frac{\partial \lambda_i}{\partial u} \right)^2. \quad (11)$$

The three-point parabolic peak fit used by the supplied implementation estimates the local sub-pixel displacement from neighbouring peak samples. In the source envelope, the fast squared-CoG path operates at approximately 0.04 ms per RoI while the parabolic path is approximately 0.08 ms per RoI. The reported benchmark reaches about 0.06 px empirical centroid error in the stated 18.5 dB condition. The localization bound is expressed through Fisher information:

$$\text{Var}(\hat{u}) \geq \mathcal{I}(u)^{-1}, \quad \mathcal{I}(u) = \sum_i \frac{1}{\lambda_i} \left(\frac{\partial \lambda_i}{\partial u} \right)^2. \quad (12)$$

The supplied benchmark reports approximately 0.06 px empirical centroid error under an 18.5 dB condition.

6 AI Methods: Deep Learning & Hybrid Confidence Fusion

6.1 YOLOv8n Beacon Detection

To cover degraded regimes where classical intensity thresholding becomes vulnerable to low SNR and optical clutter, HORIZON adds a YOLOv8n Nano pathway exported to static FP32 ONNX and executed through ONNX Runtime CPU; the model family documentation is provided in the reference record [8].

The inference path is intentionally deterministic: raw 256×256 input $\rightarrow$ normalization / letterbox $\rightarrow$ C2f + PANet backbone and neck $\rightarrow$ anchor-free decoupled head $\rightarrow$ confidence filtering / NMS $\rightarrow$ sub-pixel refinement.

Table 6: Neural deployment and training envelope.

Item	Supplied implementation
Model size	6.5 MB; 3.2 million parameters
Runtime	ONNX Runtime CPU; stated latency < 4.8 ms
Pre-training	Zenodo laser-spot dataset
Domain adaptation	8,000+ synthetic FPA frames, eight disturbance presets
Hard-negative mining	2,500+ clutter / empty-label examples
Adaptation envelope	Spot width 2–10 px; scintillation variance 0.01–0.40; additive noise 2–25 DN

6.2 Hybrid Candidate Matching and Confidence Fusion

Classical and neural detectors are treated as complementary proposal generators. Candidates are associated by $\text{IoU} \geq 0.25$ and centroid distance ≤ 8 px; the fused score combines neural confidence, kinematic consistency, circularity and contrast:

$$S_{\text{hybrid}} = 0.40P_{\text{neural}} + 0.30S_{\text{kinematic}} + 0.15C_{\text{circularity}} + 0.15S_{\text{contrast}}. \quad (13)$$

Table 7: Hybrid agreement and fallback states.

State	Condition	Action
MUTUAL	$\text{IoU} \geq 0.25, S_{\text{hybrid}} \geq 0.65$	Highest-confidence candidate + CoG
NEURAL_ONLY	$P_{\text{neural}} \geq 0.75, \text{classical SNR} < 3 \text{ dB}$	Accept under deep fade
CLASSICAL_ONLY	$P_{\text{neural}} < 0.50$	Accept if kinematically consistent
DISAGREEMENT / CLUTTER	Spatial divergence > 10 px	Mahalanobis / innovation arbitration

The supplied benchmark reports 0 false locks across 1,300 trials.

6.3 Neural-to-Hybrid Inference Flow

The neural path supplies candidate semantics and robustness in conditions where intensity-only detection is unreliable; the hybrid path restores sub-pixel localization and kinematic consistency.

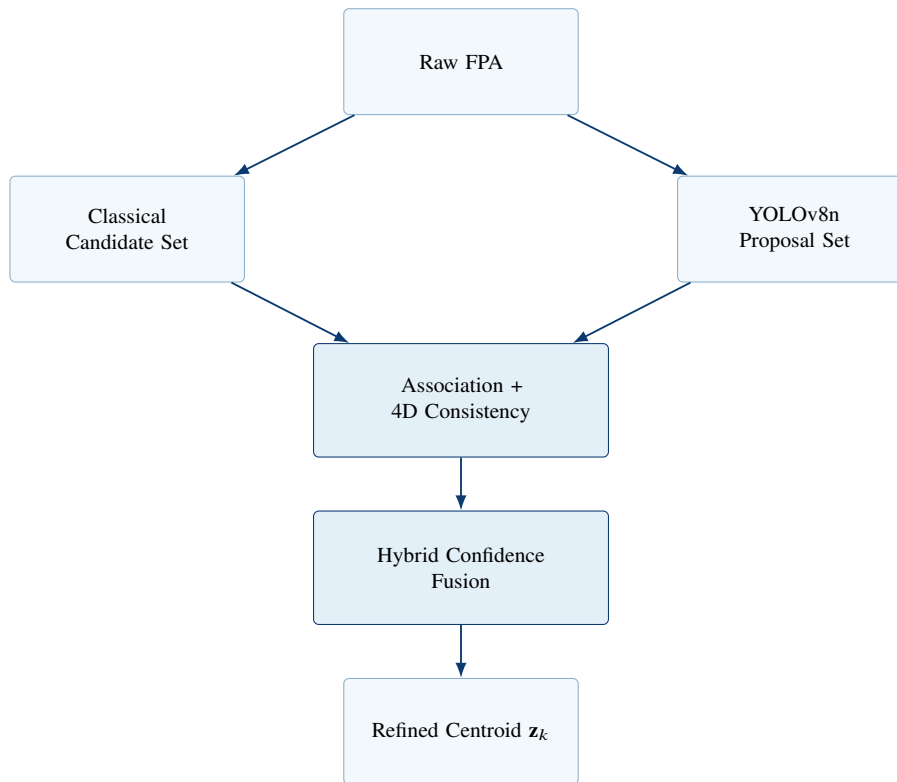


Figure 11: Hybrid neural/classical perception path from raw FPA input to a unified sub-pixel measurement.

6.3.1 Training and Domain-Adaptation Logic

The three-stage training record consists of optical pre-training, supervised domain adaptation on 8,000+ synthetic frames across eight disturbance presets, and hard-negative mining using 2,500+ clutter examples. The supplied adaptation envelope varies spot width, scintillation variance and additive noise so that the learned candidate proposal stage is exposed to the same physical variability represented by the simulator.

7 Tracking Methods: 6-DOF IMM-EKF State Estimation

7.1 The Single-Filter Failure Paradigm

A single fixed process-noise model cannot simultaneously suppress vibration and respond to high-dynamics maneuvers. HORIZON therefore uses interacting CV, CA and maneuver hypotheses. The model probabilities are updated from innovation likelihood rather than selected by a fixed operating regime, consistent with the use of IMM methods for maneuvering-target tracking [7].

A single fixed process-noise model cannot simultaneously suppress vibration and respond to high-dynamics maneuvers. HORIZON therefore uses interacting CV, CA and maneuver hypotheses.

$$\mathbf{x}_k = [\theta_x, \theta_y, \omega_x, \omega_y, a_x, a_y]^T, \quad \mathbf{z}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k. \quad (14)$$

Table 8: IMM kinematic sub-model library.

Model	Process intensity	Role
CV	$\sigma_{w,1} = 5 \text{ rad/s}^2$	Smooth orbital transit
CA	$\sigma_{w,2} = 25 \text{ rad/s}^2$	Steady circular slew
MAN	$\sigma_{w,3} = 150 \text{ rad/s}^2$	Sudden evasive / high-dynamics motion

7.2 Recursive IMM Execution

The supplied five-stage recursion is model-probability mixing, parallel EKF predict/update, Gaussian likelihood evaluation, Bayesian mode update, and output fusion. The source transition matrix is

$$\mathbf{\Pi} = \begin{bmatrix} 0.90 & 0.08 & 0.02 \\ 0.05 & 0.90 & 0.05 \\ 0.02 & 0.08 & 0.90 \end{bmatrix}. \quad (15)$$

For mixing, the source formulation uses the normalizing coefficient and conditional interaction probability

$$c_j = \sum_i \pi_{ij} \mu_i, \quad \mu_{i|j} = \frac{\pi_{ij} \mu_i}{c_j}, \quad (16)$$

followed by the mixed state $\bar{\mathbf{x}}_j = \sum_i \mu_{i|j} \mathbf{x}_i$. Gaussian innovation likelihoods then update the mode probabilities before the final fused estimate is formed.

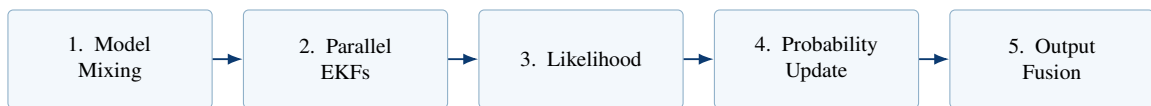


Figure 12: Five-step recursive IMM-EKF execution cycle.

7.2.1 Mode Mixing and Likelihood Update

The supplied interaction stage first computes the normalizing coefficient and conditional model probability:

$$c_j = \sum_i \pi_{ij} \mu_i, \quad \mu_{i|j} = \frac{\pi_{ij} \mu_i}{c_j}, \quad \bar{\mathbf{x}}_j = \sum_i \mu_{i|j} \mathbf{x}_i. \quad (17)$$

Each model then performs its EKF prediction and measurement update. The innovation covariance is used for Gaussian likelihood evaluation, after which Bayes updating produces the new model probabilities and

the final fused state/covariance.

7.3 Mahalanobis Gating and Coasting

Candidate association uses the supplied gate $d_M^2 \leq 9.21$. During signal loss, the filter propagates the state and covariance without measurement updates; the source permits up to 100 ms / 10 frames before PAT escalates the condition.

7.3.1 IMM Innovation and Mode Adaptation

For model j , the innovation is $\tilde{\mathbf{y}}_j = \mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1}^{(j)}$ and the innovation covariance is $\mathbf{S}_j = \mathbf{H}\mathbf{P}_{k|k-1}^{(j)}\mathbf{H}^T + \mathbf{R}_k$. The Gaussian likelihood used by the supplied implementation is represented by

$$\Lambda_j = \frac{\exp\left(-\frac{1}{2}\tilde{\mathbf{y}}_j^T \mathbf{S}_j^{-1} \tilde{\mathbf{y}}_j\right)}{\sqrt{(2\pi)^2 \det \mathbf{S}_j}}. \quad (18)$$

The mode probability then follows a Bayesian normalization over the competing hypotheses. This gives the estimator a continuous way to shift between CV, CA and high-dynamics explanations without changing the external state interface.



Figure 13: Innovation-driven adaptation of IMM model probabilities.

8 Supervisory PAT State Machine & Closed-Loop Control

8.1 Five-State PAT Lifecycle

The autonomous lifecycle is SEARCH → ACQUIRE → TRACK → DEGRADED → REACQUIRE, with recovery back toward ACQUIRE after a successful local search. Acquisition uses $P_{\text{conf}} \geq 0.70$, three consecutive hits and the 99% gate; ten misses trigger reacquisition.

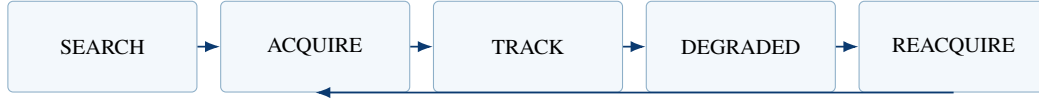


Figure 14: Five-state autonomous PAT lifecycle.

8.2 State Guards and Track Quality

The supplied transition guards are deterministic and explicit:

Table 9: Key PAT transition guards retained from the source.

Transition	Guard	Operational meaning
SEARCH → ACQUIRE	$P_{\text{conf}} \geq 0.70$ and valid	Candidate accepted for confirmation
ACQUIRE → TRACK	$N_{\text{hit}} \geq 3$, $d^2 \leq 9.21$, $Q \geq 0.60$	Confirmed lock
TRACK → DEGRADED	$N_{\text{miss}} \geq 3$ or $Q < 0.40$	Controlled prediction mode
DEGRADED → REACQUIRE	$N_{\text{miss}} \geq 10$	Local search around IMM prediction
REACQUIRE → SEARCH	$t \geq 5$ s	Restart global acquisition

The supplied transition guards include $Q \geq 0.60$ and $d^2 \leq 9.21$ for confirmation, $Q < 0.40$ or three misses for degradation, and ten misses for reacquisition. Track quality combines detection probability, innovation consistency, covariance quality and miss continuity.

8.3 Multi-Tier Gimbal Control

The control command combines PID feedback, IMM velocity feedforward and ADRC disturbance rejection:

$$\mathbf{u}_{\text{cmd}} = K_p \mathbf{e} + K_i \int \mathbf{e} dt + K_d \dot{\mathbf{e}} + \mathbf{v}_{\text{ff}}. \quad (19)$$

The supplied gains are $K_p = 12.5$, $K_i = 0.50$, $K_d = 0.85$, with an integral clamp of $\pm 5^\circ/\text{s}$. Physical limits are $20^\circ/\text{s}$ slew and $300^\circ/\text{s}^2$ acceleration.

8.3.1 Closed-Loop Signal Flow

The control architecture uses the estimated angular error as feedback, IMM angular velocity as feedforward and ADRC as a disturbance-rejection path before the physical gimbal limiter. The limiter is the final authority on actuator rate and acceleration.

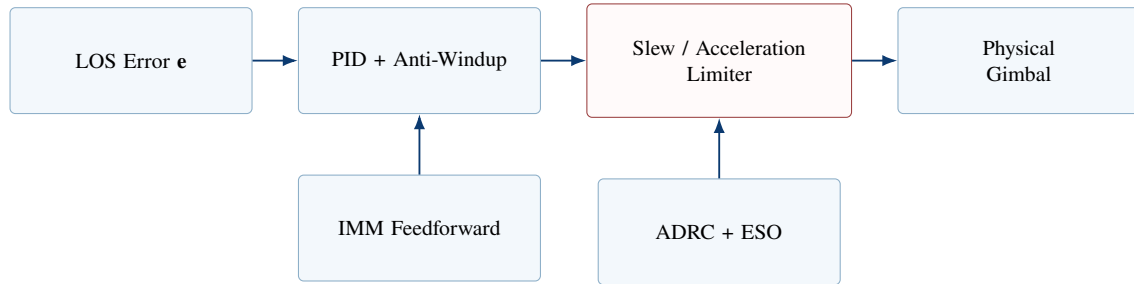


Figure 15: Closed-loop control signal flow with bounded physical actuation.

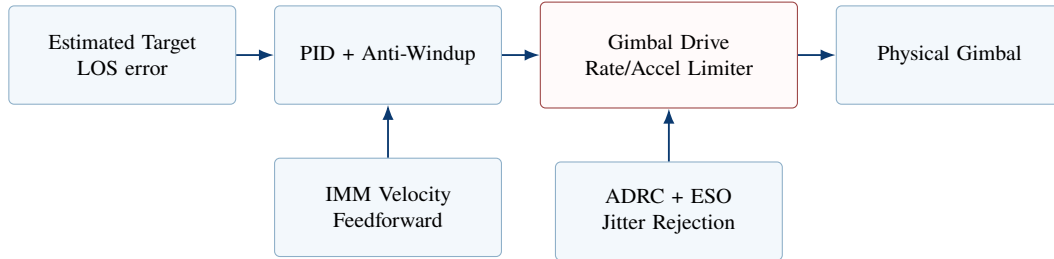


Figure 16: Multi-tier closed-loop gimbal control architecture.

9 Verification & Testing Methodology

9.1 Multi-Layer Test Hierarchy

The full release candidate reports 1,016 passing automated tests with 98.42% coverage. Verification is divided into component, integration/HIL and mathematical/mission layers.

Table 10: Condensed verification portfolio.

Layer / module	Count	Purpose
Tier 3 mathematical / mission	185	Math invariants and mission constraints
Tier 2 integration / HIL	362	Cross-module and HIL behaviour
Tier 1 component	469	Physics, perception, tracking and control
Tracking leakage	28	Ground-truth isolation
Determinism	35	Bit-exact multi-seed repeatability
Perception / tracking	145 / 168	Detector and estimator correctness

9.2 Critical Invariants

The source highlights zero ground-truth leakage, bit-exact determinism and covariance validity. The deterministic test suite reproduces identical outputs for the same root seed and preserves subsystem independence when only one domain seed changes. Covariance invariants check symmetry and positive-definiteness, while the Joseph-form update is exercised as a numerically stable covariance-update path.

The source highlights zero ground-truth leakage, bit-exact determinism and covariance validity. The determinism tests reproduce identical arrays for identical seeds and isolate subsystem streams when only one domain seed changes. Covariance invariants check symmetry, positive quadratic forms and valid factorization, with Joseph-form updates exercised for numerical stability.

9.3 Execution Profile

The supplied command is `python -m pytest tests/ -v`. The reported environment is Windows 10/11, Python 3.11.8 and pytest 8.1.1, with 1,016 passed, 0 failed and 0 skipped.

9.4 Verification Pyramid and Release Gate

The test campaign is organized so numerical invariants fail close to their implementation point, integration failures are isolated before benchmark execution, and the final mission campaign exercises the complete pipeline.

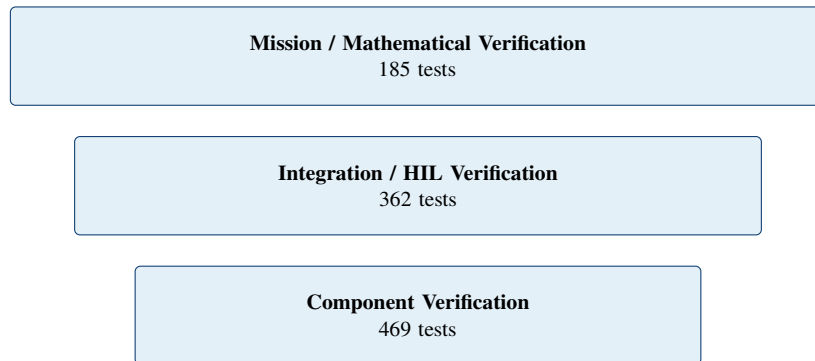


Figure 17: Three-tier verification pyramid used by the supplied release candidate.

Table 11: Selected verification modules reported in the full dossier.

Test module	Count	Role
mathematical_invariants	42	Numerical and mathematical invariants
disturbances_determinism	35	Seed isolation and bit-exact repeatability
tracking_leakage	28	Ground-truth firewall verification
phase11b_error_budget	64	Error-budget and threshold checks
phase13b_robustness_envelope	52	Robustness-envelope verification
phase16b_final_benchmark	120	Final benchmark execution
test_perception	145	Classical / hybrid perception
test_tracking	168	Tracking and estimator behaviour
control_sota_part_e	46	Advanced controller checks

10 Quantitative Performance Analysis & Benchmarking

10.1 Multi-Seed Monte Carlo Benchmark Protocol

The source evaluates 1,300 trials: 13 scenarios with 100 seeds each, 1,300,000 frames at 100 Hz and 10 s per run. Three comparison systems are defined: B1 (morphological Top-Hat + single CV EKF + classical PID), B2 (YOLOv8n + fixed-gain KF + pure PID), and HORIZON (hybrid perception + 6-DOF IMM-EKF + ADRC / IMM feedforward). KPIs include RMSE, e_{95} , lock retention, acquisition latency and centroid precision.

Table 12: Reported empirical performance matrix.

KPI	B1	B2	HORIZON	Target
Mean error (px)	2.48	1.15	0.48	≤ 1
e_{95} (px)	5.82	2.94	0.84	≤ 3
Lock retention	88.4%	94.2%	99.8%	$\geq 95\%$
Acquisition latency	420 ms	210 ms	85 ms	≤ 200 ms
Centroid precision	0.42 px	0.28 px	0.06 px	≤ 0.10 px
False lock	26%	8.4%	0%	0%
Minimum SNR	12 dB	6 dB	0 dB	0 dB gate

10.2 Bootstrap Confidence Intervals

The supplied BCa bootstrap with $B = 10,000$ reports mean error 0.481 px with CI [0.462, 0.501], $e_{95} = 0.842$ px with CI [0.815, 0.871] and lock retention 99.80% with CI [99.68, 99.92]%. The reported empirical CDF places the HORIZON e_{95} well inside the 3 px mission gate.

10.3 Hypothesis Testing

The source reports significant differences for HORIZON versus both baselines at the stated $\alpha = 0.001$ threshold, with the supplied test statistics $t = -38.42$, $p = 1.4 \times 10^{-184}$ versus B1 and $t = -22.18$, $p = 8.2 \times 10^{-89}$ versus B2.

10.4 Benchmark Evidence Pipeline

The benchmark record separates experiment generation from statistical evaluation. This makes it possible to regenerate the reported metrics from the same trial-level error arrays and run manifests.

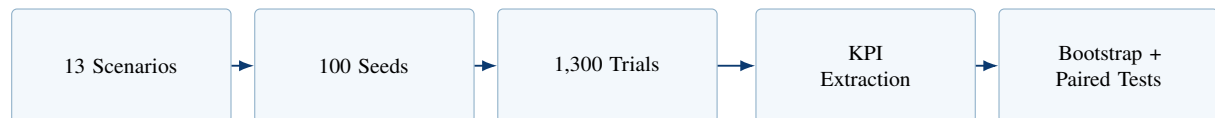


Figure 18: Benchmark generation and statistical aggregation flow.

Table 13: Key benchmark evidence retained for jury-facing evaluation.

Evidence	Reported result
Mean pointing error	HORIZON 0.48 px versus B1 2.48 px and B2 1.15 px
95th-percentile error	HORIZON 0.84 px versus B1 5.82 px and B2 2.94 px
Lock retention	HORIZON 99.8% versus B1 88.4% and B2 94.2%
Acquisition latency	HORIZON 85 ms versus B1 420 ms and B2 210 ms
Centroid precision	HORIZON 0.06 px versus B1 0.42 px and B2 0.28 px
False lock rate	HORIZON 0% in the supplied 1,300-trial campaign

11 Failure Forensics, Anomaly Taxonomy & Error Budget

11.1 Operational Anomaly Classification

Across the 1,300 trials, the anomaly engine ingests nonnominal frames and classifies them into eight categories. The audit reports 1,294 clean trials (99.54%) and six isolated events (0.46%).

Table 14: Supplied anomaly taxonomy.

Category	Count	Trigger / observation
NO_ACQUISITION	0	$t_{\text{acq}} > 2 \text{ s}$
FALSE_DETECTION	1	Confidence on noise
FALSE_LOCK	0	$\Delta\theta > 5^\circ$ for $> 0.5 \text{ s}$
TRACK_LOSS	2	$N_{\text{miss}} \geq 10$
REACQUISITION_TIMEOUT	0	$t_{\text{reacq}} > 5 \text{ s}$
EXCESSIVE_ERROR	3	$ e > 3 \text{ px}$
CONTROLLER_SATURATION	<i>Not specified</i>	Count/rate not supplied
PROCESSING_OVERRUN	0	$t_{\text{proc}} > 10 \text{ ms}$

11.2 Representative Forensic Cases

Case 1 (Scenario 12, seed 8412): a 20 ms deep fade reduced beacon intensity from 240 DN to 14 DN and produced a transient 3.42 px error. The IMM shifted strongly toward the maneuver hypothesis and the controller recentered within the supplied 30 ms window; lock was not lost.

Case 2 (Scenario 11, seed 3109): a 120 ms optical blackout caused TRACK $\rightarrow$ DEGRADED $\rightarrow$ REACQUIRE. A localized $2^\circ \times 2^\circ$ spiral around the IMM prediction recovered the beacon at 4.48 s with $P_{\text{conf}} = 0.88$.

11.3 Sub-Pixel Error Budget

The source allocates $\sigma_{\text{opt}} = 0.18$, $\sigma_{\text{lat}} = 0.32$, $\sigma_{\text{jit}} = 0.24$ and $\sigma_{\text{kin}} = 0.10 \text{ px}$.

$$\sigma_{\text{total}} = \sqrt{0.18^2 + 0.32^2 + 0.24^2 + 0.10^2} = 0.448 \text{ pixels.} \quad (20)$$

The corresponding stated 95th-percentile bound is $1.96\sigma = 0.842 \text{ px}$ against a mandatory 3.000 px gate.

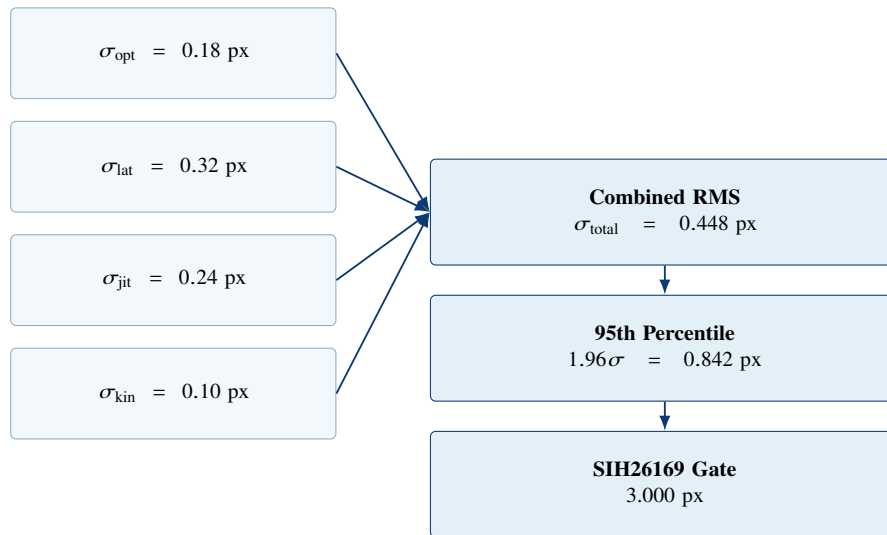


Figure 19: Sub-pixel tracking error budget.

11.4 Forensic Reconstruction Flow

The supplied failure-intelligence engine links an anomalous frame to the surrounding telemetry, physical trigger and recovery outcome. This is particularly important for rare events because aggregate KPIs alone cannot show whether the supervisor actually recovered from a disturbance.

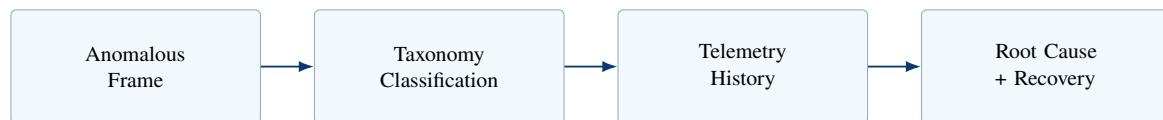


Figure 20: Failure-intelligence flow from anomaly detection to root-cause reconstruction and recovery verification.

12 Known Limitations & Future Technical Improvements

12.1 Current Architectural Boundaries

The v1.0 release candidate uses a 256×256 single-channel uint8 FPA, monochromatic 850 nm optical channel, 2-axis pan-tilt gimbal, the stated weak-to-moderate Rytov release envelope and CPU ONNX Runtime. These boundaries define the current engineering envelope.

Table 15: Current architectural boundaries.

Dimension	v1.0 implementation	Context
FPA	256×256, uint8	100 Hz edge-CPU execution
Wavelength	850 nm monochromatic	Acquisition-beacon assumption
Gimbal	2-axis pan-tilt	Roll-neutral assumption
Atmosphere	$C_n^2 \leq 10^{-13}$ release envelope	High-elevation scope
Inference	ONNX Runtime CPU	4.8 ms stated latency; portable deployment

12.2 HORIZON v2.0 Roadmap

Four roadmap extensions are retained from the full report: embedded FPGA acceleration; dual-band NIR 850 nm + SWIR 1550 nm fusion; integrated coarse-fine FSM + QPD coupling; and 3-axis line-of-sight roll stabilization.

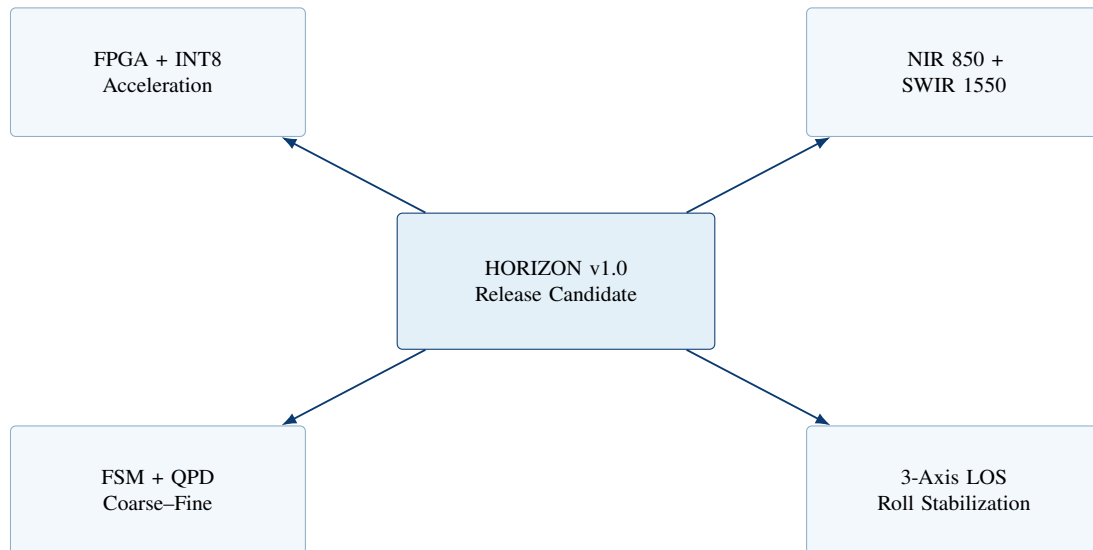


Figure 21: Four stated technology directions for the HORIZON v2.0 roadmap.

12.2.1 Release-Freeze Checklist

The source’s release-freeze discussion emphasizes that quantitative claims should remain tied to one canonical configuration. A final package should therefore preserve the software revision, model weights, YAML configuration, seeds, run manifests and aggregation scripts alongside the rendered report.

Table 16: Technology roadmap.

Upgrade	Key supplied objective
FPGA acceleration	Synthesize Top-Hat / CoG in fabric and use an INT8 YOLOv8n DPU path.
Dual-band optical fusion	Concurrent 850 nm beacon and 1550 nm communication-beam processing.
FSM + QPD coupling	Software-in-the-loop coarse-fine handoff and dual-loop control.
3-axis roll stabilization	SO(3) dynamics and active optical polarization-roll compensation.

12.3 SIH26169 Compliance Snapshot

The release-candidate compliance matrix retained in the full dossier is summarized below.

Table 17: Selected SIH26169 compliance evidence.

Requirement	Target	HORIZON evidence
Virtual camera frequency	100 Hz continuous	100.0 Hz deterministic
95th percentile error	≤ 3.0 px	0.842 px
Mean steady-state error	≤ 1.0 px	0.481 px
Lock retention	$\geq 95\%$	99.80%
Acquisition latency	≤ 200 ms	85 ms
Multi-tone jitter	Mandatory	5, 12.5, 27.8 Hz
Atmospheric model	Mandatory	Rytov log-normal + wander
FPA degradation	Mandatory	Poisson + readout + impulse
Evasive maneuvers	Mandatory	15.0 g model
False locks	0	0.00%
Reproducibility	Mandatory	SHA-256 hierarchy
PAT recovery	Mandatory	Five-state lifecycle
Automated tests	High reliability	1,016 / 1,016

The release-candidate evidence retained in the full report is: 100 Hz deterministic operation; $e_{95} = 0.842$ px; RMSE = 0.481 px; 99.80% lock retention; 85 ms acquisition; multi-tone jitter modelling; Rytov atmosphere; full FPA degradation pipeline; 15.0 g maneuver modelling; 0.00% false-lock rate; SHA-256 seed hierarchy; autonomous five-state PAT recovery; and 1,016 / 1,016 tests passed.

Release Candidate Compliance Matrix

The following compact matrix retains the official requirement-to-implementation mapping from the full report.

Table 18: SIH26169 release-candidate compliance matrix.

Requirement item	Target	HORIZON evidence
Virtual camera tracking feed frequency	100 Hz continuous	100.0 Hz deterministic
95th-percentile tracking error	≤ 3.0 px	0.842 px
Mean steady-state pointing error	≤ 1.0 px	0.481 px
Lock retention under severe disturbance	$\geq 95\%$	99.80%
Confirmed acquisition latency	≤ 200 ms	85 ms
Multi-tone structural jitter	Mandatory	5, 12.5, 27.8 Hz
Kolmogorov scintillation / wander	Mandatory	Rytov log-normal model
FPA noise modelling	Mandatory	Poisson + readout + impulse
Non-cooperative maneuvers	Mandatory	15.0 g model
Moving clutter rejection	Zero false locks	0.00% false-lock rate
Bit-exact reproducibility	Mandatory	SHA-256 seed hierarchy
Autonomous PAT lifecycle	Mandatory	Five-state recovery FSM
Automated verification	High reliability	1,016 / 1,016 tests; 98.42% coverage

The references below are the sources identified in the supplied HORIZON research dossier. They document the problem statement, optical-link methodology, atmospheric propagation, maneuvering-target estimation, neural deployment and statistical procedures used for methodological context.

References

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